

A Thematic Layer for CLAUDETTE: Separating the Cost of Multi-Label Annotation from the Effect of Taxonomy Granularity

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Abstract. The CLAUDETTE/UNFAIR-ToS corpus is the reference benchmark for unfair clauses in online terms of service, but it records only whether a sentence is unfair, never what the clause is about. We add a thematic layer to its 50 contracts: 9,414 sentences, each labelled with one or more of 20 legal themes independently by three trained annotators, and by four large language models on the same sentences. The layer separates two effects that the reliability literature tends to conflate. Letting annotators assign several themes lowers agreement, by much the same amount whatever the number of categories. Merging the taxonomy from 20 themes into 11 recovers an equivalent amount, and the gain holds on contracts held out from the design of the merges. Merging is constrained, however: the taxonomy that agreement alone would select groups clauses that consumer protection law treats differently, at more than twice the cost in downstream signal. A fine-tuned legal encoder outperforms every language-model judge, while remaining clearly below the human ceiling. We release the annotations with their individual votes, the code-book, the protocol and the annotation platform.

Keywords. unfair terms of service, CLAUDETTE, multi-label annotation, inter-annotator agreement, LLM-as-a-judge, legal language models, annotation platform

1. Introduction

CLAUDETTE records whether a sentence in a consumer contract is potentially unfair; it records nothing about what the clause is about [1,2]. Where clause topics do exist, they come without chance-corrected agreement [3,4], so the reliability of the task itself is unknown.

We add a thematic layer to the reference benchmark. All 50 contracts and 9,414 sentences of CLAUDETTE are labelled with one or more of 20 legal themes, independently

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by three trained annotators, and separately by four large language models prompted with the same vocabulary on the same sentences.

The layer settles a question that the reliability literature leaves open. Agreement is known to fall when annotators may assign several labels, and to rise when categories are merged [5,6]; what is not known is whether these are one effect or two, since they are seldom measured on the same sentences. Here they are. The multi-label format costs 0.073 of chance-corrected agreement and varies by only 0.012 across four granularities, while merging 20 themes into 11 recovers 0.067 — a gain that replicates on 17 contracts held out from the design of the merges. Consolidation is therefore a real lever on reliability, but not an unconstrained one: the taxonomy that agreement alone would select merges clauses across unfairness strata, and loses 13.2% of the downstream unfairness signal to buy 0.012 of extra agreement.

We contribute: (i) a thematic layer over CLAUDETTE, multi-label and triply annotated, published with its disagreements and the platform that produced it; (ii) a separation of the cost of the multi-label *format* from the effect of taxonomy *granularity*, measured on the same sentences and replicated out of sample; and (iii) a legal constraint on consolidation — merges must not cross unfairness strata — together with the price of violating it.

2. Background and Related Work

Detecting potentially unfair clauses in terms of service is well established: CLAUDETTE and its successors detect and explain unfairness in English and across languages [1,2,7,8,9,10], and the task is part of standard legal benchmarks [11], on which compact domain-specific encoders remain competitive with general-purpose large models [12,13,14,15]. Language models are increasingly used as semantic annotators of legal text, with documented gains in cost and risks of anchoring and circularity [16,17,18,19,20], and, as judges, with known divergence from human evaluators [21,22,23]. On the measurement side, chance-corrected agreement for set-valued annotation is a mature subject [24,25,26,6,27], and a meta-analysis of annotation studies identifies the number of categories as one of the strongest determinants of agreement [5].

What is missing is a reliable, measured and published thematic layer on the reference benchmark itself. Contract-topic resources adjacent to ours are either single-annotated [3], defined for a different document type [28,29,30], or reported without chance-corrected agreement [4,31]; none places human annotators and language models on the same vocabulary and the same sentences. Because format and granularity are almost never varied on the same data, it is also not known whether the agreement penalty attributed to multi-label annotation belongs to the format or to the label set — a question [5] raises and leaves open.

3. The Thematic Layer and Its Construction

The layer. We annotate all 50 English contracts of CLAUDETTE [1]: 9,414 sentences, each labelled with one primary theme and zero or more secondary themes drawn from a 20-theme legal vocabulary (T20). Every sentence was labelled independently by each of

three trained annotators working from a written codebook, and, separately, by four large language models from three providers (Anthropic, OpenAI and Mistral AI; Table 2), each run through its vendor’s agent interface at default settings, prompted once with the same vocabulary on the same sentences between May and August 2026; their outputs were stored and never regenerated. Individual votes are retained rather than collapsed into a consensus [32,33,34].

Construction. Annotators did not post-edit the output of one model. The four judges labelled every sentence independently, in isolated sessions with no access to one another’s outputs or to any log of the campaign, and a deterministic engine sorted each sentence into one of five tiers on inter-judge agreement alone, from unanimous to fully split. Annotators, who could not see one another’s work, saw that tier and the competing candidates but selected from the full 20-theme vocabulary, and frequently did: 34.5% of gold labels match none of the four proposals, rising to 58.7% on the sentences annotators themselves disputed. The gold standard is built by a three-tier cascade: 46.8% of sentences are unanimous, 48.3% carry a majority of at least two of three, and the remaining 4.9% (462 sentences) were arbitrated by a human committee — the fraction of the corpus on which three trained annotators, given the same codebook, cannot converge. Crucially, *model outputs are never parties to the gold*: resolution is strictly inter-annotator, and the judges are held out as an object of evaluation rather than a contributor to the reference [23].

T14 and T11 were designed under four criteria — confusability, reliability, contractual function and support — and one constraint fixed in advance: no merge joins themes whose observed unfairness rates fall in opposite strata. The merges were reviewed by the legal scholars among the authors; T10 breaks that constraint by construction and is reported as a counterfactual.

Measurement. We report Krippendorff’s α under two distances [24,26]: the MASI distance for set-valued labels [25], and the nominal distance on primary themes alone. The gap between them is what multi-label annotation costs in reliability [6]. Coarser taxonomies (T14, T11, T10) are *projections* of T20 applied at read time, so every comparison is paired by construction; confidence intervals come from bootstrap resampling over documents, never sentences. The merges were designed on 33 contracts and validated on the 17 held out from that design, and the human ceiling is estimated leave-one-annotator-out against a gold to which the scored annotator did not contribute [35].

4. Results

The cost of the multi-label format. Scoring secondary themes lowers agreement from a nominal α of 0.732 to an α -MASI of 0.658: the multi-label format costs 0.073 [0.061, 0.087] of chance-corrected agreement. Across the four granularities it stays between 0.063 and 0.075, without decreasing monotonically as categories are merged (Table 1); its whole range, 0.012, is five times smaller than the agreement gain consolidation yields below.

The effect of granularity. Consolidating the taxonomy recovers agreement of the same order as the format costs. Merging T20 into T11 raises α -MASI by 0.067 [0.058, 0.077], carrying the layer across the 0.667 threshold conventionally used for tentative conclu-

Table 1. Reliability and downstream signal by taxonomy, on all 50 contracts (9,414 sentences, three annotators). The gap between α -MASI and nominal α is the cost of the multi-label format. Differences against T20 are paired by document; 95% confidence intervals from 1,000 document-level bootstrap resamples. Signal loss is the relative drop in leave-one-document-out average precision of $P(\text{unfair} \mid \text{theme combination})$, scored against the CLAUDETTE unfairness labels, which are never projected. T10 merges across unfairness strata; T14 and T11 do not.

Taxonomy	Classes	α -MASI	Nominal α	Δ vs. T20 [95% CI]	Signal loss
T20	20	0.658	0.732	—	—
T14	14	0.693	0.768	+0.035 [0.030, 0.042]	6.4%
T11	11	0.725	0.794	+0.067 [0.058, 0.077]	5.9%
T10	10	0.737	0.800	+0.078 [0.068, 0.090]	13.2%

Table 2. Agreement with the frozen gold standard, for all three kinds of system, measured by the same procedure, under the annotated taxonomy (T20) and the consolidated one (T11). Human agreement is leave-one-annotator-out, so that each annotator is scored against a gold to which they did not contribute; the encoder is evaluated 5-fold by document; judge labels were produced once and stored. Judges are ordered by agreement; their ranking is the same under both taxonomies. Reported, not demonstrated: see Section 4.

System	κ		Accuracy	
	T20	T11	T20	T11
Human annotators	0.859	—	0.871	—
Legal-BERT, fine-tuned	0.695	0.720	—	—
Judge FABLE (Claude Fable 5)	0.581	0.591	0.601	0.629
Judge CLAUDE (Claude Opus 4.7)	0.515	0.531	0.539	0.576
Judge MISTRAL (Mistral Medium 3.5)	0.327	0.353	0.358	0.413
Judge CODEX (GPT-5.5)	0.266	0.302	0.305	0.375

sions [25] — a threshold the annotation taxonomy itself does not reach. Two properties make this more than a mechanical artefact of coarser categories. First, the gain *replicates out of sample*: the 17 contracts held out from the design of the merges yield +0.069, against +0.066 on the 33 contracts used to design them. Second, T11 *dominates* the intermediate T14: it gains twice as much agreement (+0.067 against +0.035) while losing *less* downstream signal (5.9% against 6.4%). Granularity is therefore a genuine lever on reliability, and not merely a redistribution of the same disagreements over fewer categories.

The legal constraint on consolidation. Consolidation cannot be driven by agreement alone. T10, obtained by merging along measured confusion clusters without regard to the legal status of the clauses, buys a further 0.012 of α -MASI over T11 — and loses 13.2% of the unfairness signal instead of 5.9%, a yield of 0.59 points of agreement per unit of signal against 1.13 for T11. The merge responsible joins two themes whose observed unfairness rates fall in opposite strata, 0.385 against 0.039 — a distinction consumer protection law draws [36,37] and that the agreement signal does not express. A purely statistical consolidation selects exactly this taxonomy; the legal constraint rules it out: reliability gains and downstream utility are not interchangeable, and the boundary between them is drawn by law, not by data.

Reported: the human ceiling, the judges and a fine-tuned encoder. Against the frozen gold, trained annotators reach $\kappa = 0.859$ in leave-one-annotator-out, the best of the four

LLM judges reaches 0.581, and a fine-tuned Legal-BERT reaches 0.720 on T11 (Table 2). The judges span κ from 0.27 to 0.58 and keep their ranking under T11, gaining only 0.01–0.04 where the annotators recover 0.067; the two strongest are nearly redundant ($\kappa = 0.80$ between them, above any human pair), so the benchmark holds three distinct behaviours rather than four. Reported rather than demonstrated (Section 5), their ordering is nonetheless unambiguous: a compact, domain-specific encoder outperforms every judge while remaining well below the human ceiling.

5. Discussion and Limitations

Two consequences follow for annotation practice. First, the reliability cost of multi-label annotation need not be paid by abandoning multi-label annotation: it can be recovered by consolidating the taxonomy. Second, that consolidation cannot be delegated to the agreement signal: a purely data-driven procedure selects the taxonomy that collapses the legal distinction the downstream task depends on, so the criterion has to come from outside the annotation statistics [36,37]. For the evaluation of language models, the gap between the human ceiling and the best judge (Table 2) is large enough that judges are not substitutable for trained annotators on this task [23,22]; a compact domain-specific encoder closes much, though not all, of that gap.

Limitations. The four judges we evaluate are the four that assisted annotation, so the benchmark is not free of circularity [20]: the engine favoured none of them and a third of gold labels match no proposal, but an error shared by all four could survive. FABLE alone gains accuracy where the annotators split (0.607 on unanimous sentences, 0.636 on majority-decided ones; CLAUDE 0.578 and 0.543), the signature of anchoring rather than competence. Excluding it leaves CLAUDE at $\kappa = 0.515$, which widens the gap to the human ceiling rather than narrowing it. The layer covers one legal system and one language, and the stored judge labels measure a historical output rather than the current state of the art. The fine-tuned encoder is a single model with early stopping on the test fold, making its reported agreement mildly optimistic. Finally, the aggregated gold carries only 1.04 labels per sentence, so a multi-label score computed on it would not measure the difficulty of the task as annotated; the multi-label cost reported here is measured on individual votes, where the secondary themes survive.

6. Conclusion and Availability

A thematic layer over CLAUDETTE makes it possible to ask what a clause is about, not only whether it is unfair, and to measure how reliably the question can be answered: the cost of the format and the effect of granularity are separable and of different orders, and the consolidation agreement alone would favour is ruled out by a legal constraint no statistical criterion expresses.

Availability. We release the layer with its individual votes, arbitration trail and frozen gold standard, the codebook, the protocol, the exact prompts given to the four judges, the frozen projections with the fingerprint under which every figure was computed, and the annotation platform [38,39,40]: data and source at <https://github.com/elazharjebbari/claire-studio>, running instance at <https://pactiva.legal>.

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